CREST AWARDS | GOLD

This CREST Gold Award is presented to:

Aarav Jain

for the project

Smart Irrigation System

13/06/2025

Step by Step School

Heather King

Dr Heather King
Vice-President for Education, British Science Association

Managed by BRITISH SCIENCE ASSOCIATION Supported by UK Research and Innovation Urenco

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Knowledge Prism

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To Whomsoever It May Concern

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As the Director of the Indigo Knowledge Prism Foundation and Founder of Learnovate, I have spent years working at the intersection of education, innovation, and community development. Through curriculum initiatives, experiential learning programs, and grassroots technology projects, I have had the privilege of mentoring and evaluating hundreds of young students. It is therefore with great enthusiasm that I recommend Aarav A. Jain, a remarkable young innovator whose rare combination of technical curiosity, community leadership, and social consciousness distinguishes him far beyond his peers.

I first came to know Aarav through LinkedIn. Unlike many students who reach out seeking short-term volunteering opportunities, Aarav approached me with a clear purpose. He spoke about his longstanding fascination with water systems, shaped in part by years spent as a swimmer and by his growing concern about the sustainability challenges surrounding water use in India. What particularly stood out was that his interest extended beyond environmental rhetoric. He was actively seeking to understand how technology could be deployed to address real-world problems.

Our conversation revealed a young man who had already begun this journey. During a visit to agricultural communities in Assam, Aarav had spent time observing irrigation practices and interacting with farmers. What he witnessed left a lasting impression. Despite increasing concerns around water scarcity, large volumes of water were still being lost through inefficient irrigation systems and manual monitoring practices. Rather than treating this as an isolated observation, Aarav began investigating whether affordable automation could help farmers conserve water while reducing the burden of daily irrigation management. That inquiry ultimately evolved into the concept of a low-cost smart irrigation system.

Around the same time, Aarav had established himself as a committed mentor and educator. Whether guiding younger students within his school community, supporting participants through STEM outreach programs, or teaching underserved children through educational initiatives, he consistently demonstrated a belief that technology becomes most meaningful when it empowers others. This philosophy naturally aligned with the mission of our Village Research Lab initiative, which brings together students from rural communities to solve local challenges using emerging technologies.

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Over a period of 2 weeks, Aarav worked closely with 15 students from farm families in a village located approximately 30 kilometers outside Gurugram. Rather than building the irrigation system himself and presenting it as a finished product, Aarav chose a far more challenging approach: teaching the students to design, assemble, and deploy the systems themselves.

Under his guidance, the students learned to integrate soil moisture sensors with ESP32 microcontrollers, relay-controlled pumps, and OLED displays to create autonomous irrigation systems capable of monitoring soil conditions and regulating water usage in real time. What impressed me most was not the sophistication of the technology, but Aarav's approach to teaching. He consistently encouraged students to investigate, experiment, and troubleshoot independently rather than relying on him for answers. In doing so, he transformed technical instruction into genuine problem-solving.

As the project evolved, Aarav continually pushed the students to think beyond engineering specifications and consider broader questions of impact. He encouraged discussions around water conservation and long-term sustainability. His ability to connect technical design with practical implementation reflected a maturity that is rarely seen at his age.

The results were significant. By the conclusion of the program, the students had successfully deployed smart irrigation systems across 15 family-owned farms. Early observations indicated approximately a 30% reduction in water consumption compared to conventional irrigation practices. Equally important was the confidence the students gained through the process. Many began viewing technology not as something distant or inaccessible, but as a tool they could use to improve conditions within their own communities.

Over the years, I have worked with many talented young people, but very few have demonstrated Aarav's combination of intellectual curiosity, leadership, and commitment to service. He brings seriousness to his work without losing humility, and ambition without losing sight of the communities he hopes to serve. I am confident that wherever Aarav chooses to study and contribute next, he will continue developing solutions that are both innovative and meaningful. He is the kind of individual who does not wait for opportunities to create impact—he actively seeks them out and follows through with exceptional dedication.

I wholeheartedly recommend Aarav A. Jain with my highest enthusiasm.

Sincerely,
Sachin Kumar
Director
Indigo Knowledge Prism Foundation